Accor Assessment

Q1 - Compute & Architecture Design

Compute & Architecture Design a containerized infrastructure on AWS EKS. Your solution must demonstrate how the application remains available and responsive during extreme load events and infrastructure failures.

In order to design a full scale high availablity system we need to understand the complete user journey, nu
flown every week, every month, during sale and promotions to handle and process the user request traffic.
Considering infrastructer load distribution, network, security, api server, kubernetes

```
User request
     |
  https request (TLS)
     |
■ Route53 (DNS resolution)
     |
■ Cloud Front
     |
    Application Web Firewall
     |
 API gateway
     |
  Application load balancer
     |
    Ingress
     |
     Service
     |
    EKS
     |
-----------------------------------
|      |          |          |
API   etcd     Controller Scheduler
Server           Manager
-------------------------------------
                |
              Pods on Multi Availablity Zones for high availablity and container runtime
             Pod 1 |            Pod 4 | AZ2      Pod 6 AZ3 - Node 3
             Pob 2 |-Node 1AZ1  Pod 5 | Node2
             Pod 3 |
                |
             Redis cache (for faster transactions to store static data)
                |
               RDS on AZ1 (primary)
                -Standby RDS on AZ2
                      |
```

Wrker node EC2 Instances
Kubelet
Nodes
Pods
Container Runtime
Containerd

HPA
Karpenter
Cluster Autoscaler

PDB (pod disruptin budget) to be defined for high availablity which defines the minavailable:2 pods should

Liveness Probe - If the applicaiion is crashing and kubernetes restarts the container
Readiness probe - Checks if DB,nodes, pods are ready to take the traffic, if its failed, pods are removed f

AWF - Allows only those requests whch are set to be allowed through firewall to ALB 443
It absorbs excess traffic protects SQL ingestions like multiple null pointers, Ddos attacks. Makes the lay

ALB - Distributes the traffic requests

Request hits the API gateway on 8080 port where the gateway authenticates the request
on the exposed api

Ingress toute the request from api gateway to the service based on the endpoint like
accor.com/search
accor.com/availiblity
accor.com/loyalty-points
accor.com/payment

Service distributes service request to the pods like a load balancer

HPA - Keeps a check on the CPU Target, lets say if we define threshold at 75%, HPA based on /metrics inform

Controller manager : Creates new replicas set and it creates new pods based on desired state vs actual stat

Schedule assigns the pods to the nodes based on the load and availablity and based on Pod Affinity rules wh
defined atleast 1 pod to be running on each node.

Cluster autoscaler or karpenter spins us additional nodes and places new pods whereas karpenter is faster t

Kubelet : Its a node agent than run the pod by calling container runtime where containerd will be pulling i

Multi availablity zone which is highly required for higher availablity.
and due to cross region read replicas the availablity remains high. on Pods, Nodes, RDS, S3 registry, DR

Define RTO (recovery time obejctive) - 10minutes
and Recovery Point Objective (1mins) to sync database read replicas before any failure

Monitoring and Observability
Cloud Watch
Prometheus
Grafana
Datadog

Kafka is used for queue which uses broker which communicates and processes between producer and consumer us

During extreme traffic spikes, Redis reduces database load by querying static data and not hitting the data
HPA automatically scales application pods by creating replicaSet
Cluster Autoscaler or Karpenter provisions additional worker nodes if the node dies or CPU,Memory utilisati
During infrastructure failures, EKS reschedules pods to healthy nodes,
Multi-AZ deployment ensures application availability, and RDS Multi-AZ provides automatic database failover

Q2 - Scalability Strategy

Scalability Strategy The system must automatically handle the 10x traffic spike scenario without manual intervention. Define your strategy for scaling both the application and the underlying infrastructure

Answer - Assuming normal rate of request is 100 requests/second and it spikes to 1000 request per seconds

Use predictive modelling based on the previous data by adding 20% additional count

We will have to assign Horizontal Pod Autoscaler which senses the CPU utilisation based on the targetcpuuti
where the controller manager detects the change in desired state vs actual state and create pods

If 6 pods are running to process 100 req/sec, HPA will scale it to 60 pods
By increasing the number of pods the node resource utilisation will also spike and pods might face OOM issu
Cluster autoscaler or karpenter will increase the nodes and add it to the cluster where the scheduler will

Define CPU & Memory request vs limits

Redis cache - It stores the static information like loyalty points,search filters, redemption points, addre
to avoid latency and if not available it goes to the database
Multi Availablity zones where the Nodes, Pods, RDS, S3 will have cross region read replicas.
Read replica can be used to distribute the DB request among primary/master and slave (read replica)

CDN (Content drivern network)/Cloud Front - Ddos attacks, cacheing api,images, redemption points which redu

AWF - filters bot requests on specific api, protects SQL injections

Kafka - Queues are loaded where broker processes the queue between producer to consumer and fastens. It abs

Q3 - Security & Networking

Security & Networking Design a network and security architecture that protects customer data and infrastructure integrity. Apply "Least Privilege" and "Defense in Depth" principles to the entire stack.

Answer - Create a Virtual private cloud in AWS and define public and private subnets
Public subnets that are exposed to the internet such as AWF and ALB

Private subnets for EKS cluster, Redis, RDS as the internal data and user data should not be exposed to int

https TLS connections
Encrypted TCP packet

Route53 which resolves DNS

Cloudfront, Application web firewall

API gateway for authentication of requests and services

IAM for authentication and authorisation where Users, Rules, Groups and Policies

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  Application load balancer
     |
    ingress
     |
     Service
     |
    EKS
     |
    Pods
■  |
■ Redis
■   |
■   RDS encrypted to access the database
■
■
 IRSA will define the access for pods
```

Pod A - Read loyalty_points

IAM for authentication and authorisation where Users, Rules, Groups and Policies

Groups - Lets say developer and QA can have only READ access in PROD,

operations support and SRE can have only Admin access

Network rules allow traffic only from 443 and 8080

Adding multiple security layers where if AWF is bypassed, we got cloud front, ALB, IAM, IRSA,TLS defines de

Least privilege is when IAM IRSA RBAC■ Security groups are defined where to limit pods access
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Q4 - Reliability & Observability

Reliability & Observability Define how you will measure the health of the service and how the system will recover from failures (e.g., AZ outage, bad deployment) with minimal impact to the customer.

Answer - Define Error rate,SLA,SLO and SLI

SLO is the target error rate such as 99.5%
Maintain the SLI at 99.5 and above such as 99.8%
SLA is the agreement with customer where we are charged with penalties if crossed the threshold

Configure Cloudwatch using Alert Manager
Prometheus that collects the data from exposed /metric
Grafana where data is displayed in graphical format
Datadog

We configure this to capture CPU, Memory, latency, DB, pods, Nodes, Availablity zones, logs, Traces, APM,RU

We can use helm to install this on the agents

helm install datadog-agent/datadog/datadoghq

Define targetutilisation

As soon as the resource utilisation crossed the threshold, Datadog/Grafana/Cloudwatch alerts the On-call, s
Based on the alert we check the CPU and memory utilisation along with Network and disk latency.

Readiness probe and liveness probe should be running and not failed
Liveness Probe - If the applicaiion is crashing and kubernetes restarts the container
Readiness probe - Checks if DB,nodes, pods are ready to take the traffic, if its failed, pods are removed f

Reduce toil by automating the KPI service success/failure rate as a dashboard report instead of sending it

If AZ1 fails, pods run on AZ2 and AZ3

If Node A running beyond CPU threshold HPA will add pods to healthy nodes to reduce CPU utilisation

If all the nodes crossed threshold, Karpenter add new nodes to the cluster and assign pods to it

Add rollback procedures and rollout stratergies.

kubectl rollout status deployment.yaml
kubectl rollout undo deployment.yaml

Use of Canary or Blue- Green deployment methods to check system performance and utilisation and gradually i
In Blue- Green we push the deployment in stand by server and make it primary

Q5 - Operations

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Operations
• Day 2 Operations: Propose strategies to minimize operational toil.
• Task Assignment: Outline how the implementation tasks for this project should be
assigned across a team of 3 engineers (1 Senior, 2 Juniors).


In order to reduce the operational toil we need to understand the scope of reducing the manual efforts. Con

We need to break it into 4 sections

IAC - Terraform
Deployments - Kubernetes and/or ArgoCD
Monitoring and observability - Datadog/Cloud Watch/Grafana

Find the opportunties of automating the manual efforts

Work with product, development engineering and IT ops teams to understand the manual efforts and automate t

Infrastructure changes such as adding VPC,DB servers, Nodes to cluster, Patching updates, Adding rules to s
automated using terraform

Deployments : Pod scaling, Node scaling, Restart containers, Restart the pod, failovers to Availablity zone
can be reduced using HPA (Horizontal Pod Autoscaler), Karpenter or cluster autoscaler, failovers using kube
Also once the code is pushed to github with a PR raised to running CI/CD pipelines running build, testing,
kubectl apply -f deployment/payment can be automated using ArgoCD

Monitoring and observability: Understanding the end to end architecture and infrastructure of the applicati
capture CPU, Memory, Disk, Latency, Nodes health, Pod health of the infra resources.
Use endpoints flow to create traces, request rates, error rates,error logs, APM

Automate alerting based on the priority of the incident to slack,chat space,email, xMatters, On call.

Setup dashboards based on business and operations KPI to enhance the system better based on error percentag

Automate health check reports of infra resources bi-weekly to engineering and management.

Automate synthetic monitors and flap test that logs in with defined credentials, tests the application func
functtionality fails.

Reduce noisy alerts

Integrating Bits Ai or Resolve Ai we can introduce to triage alerts, identify root causes, and recommend re


Explain the entire flow and gaps to the team, take their ideas into considerations and based on the require

Senior Engineer will be understanding the architecture, checking the existing terraform into automating IAC
engineering teams, mentoring juniors.

Juniors Engineer 1 will be writing/updating deployment.yaml scripts to kubernetes and deployment and also u

Junior Engineer 2 will be assigned with Monitoring and Observability .

As a lead i will be working with Product, development, support, stake holders to understand the pain point,
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